22/7 H/w

1) Java generates Hash Code for every object. Find out what is the Hash code in java & the purpose of generating the Hash Code.

→ Hash Code :-

A Hash code is an integer value that Java's hashCode() method computes for an object. Every object (since it inherits from Object) has a default hashCode() method, which typically derives the value from object's memory address - though classes can (and often do) override it to base the value on object's actual field contents instead.

Syntax :-

```
Object obj = new Object();
System.out.println(obj.hashCode());
  // e.g.; 366712642
```

Purpose of Generating Hash Code:-

1) **Fast searching in collections:-** HashMap, HashSet, and Hashtable use hash codes to decide which "bucket" an object goes into. This lets Java find, insert, or check objects almost instantly (close to O(1)) instead of scanning through every element one by one.

2) **Works with equals():-** Java follows a rule if two objects are equal (equals() returns true), their hash codes must also be equal. Because of this, whenever you override equals(), you should also override hashCode() - otherwise things like HashMap may fail to find a key you already stored.

3) **Quick Comparision before full check-** Comparing two hash codes is much cheaper than comparing entire objects. Java uses hash code as a first-level filter, then confirms with equals() if needed.